

LECTURE 3: INFERENCE FOR NORMAL MEANS

Key concepts:

- Parameterizations: standard deviation, variance and precision
- Posterior mean precision-weighted average
- Posterior precision = Prior precision + $n \cdot$ Data Precision
- Prior and posterior predictive distributions
- Improper priors
- Jeffreys' prior

1. INTRODUCTION

In these notes, we review inference for the mean of a normal random variable when the variance is known. It is, of course, unrealistic to treat the variance as known but this greatly simplifies the algebra and provides important intuition about how Bayesian inference combines prior information with data. In the next two lectures, we will extend this analysis to cases where the variance is also unknown.

As in the binomial model, the posterior mean represents a compromise between the prior mean and the mean of the data. The posterior variance is always less than the prior variance and information is additive: posterior precision is the sum of prior precision and data precision (where “precision” has a precise definition, explained below). As the amount of data increases, it eventually overwhelms the prior so that the posterior distribution approaches a $\text{Normal}(\bar{y}, \sigma^2/n)$ distribution, where n is the sample size, $\bar{y}$ is the sample mean, and σ is the standard deviation of the observations.

Parameters (such as the population mean) are an analytic convenience, but generally unobservable. In assessing model adequacy, it is useful to compare observable data to their (marginal) distribution by averaging over model parameters. This is called the *prior predictive distribution*. A more stringent test is out-of-sample prediction, using the *posterior predictive distribution*. We will discuss both. Finally, we discuss uninformative prior distributions, which can be viewed as the limit of priors with precision decreasing to zero.

2. EXAMPLE: UNEMPLOYMENT INSURANCE BENEFITS

We will use a simple running example in this chapter, based on Dubin and Rivers, “Experimental Estimates of the Impact of Wage Subsidies,” *Journal of Econometrics*, vol. 56 (1992), pp. 219–242.

A sample of $n = 3,781$ initial unemployment insurance (UI) claimants were offered a wage subsidy of \$500 if they were reemployed within eleven weeks of their claim. On average, these recipients received $\bar{y} = 2166$ dollars in UI benefits during their initial spell of unemployment, with a standard deviation of 351 dollars. The purpose of the analysis was to determine the effects of the

wage subsidy in terms of time unemployed, cost of UI benefits, and reemployment earnings. Here we shall focus on UI benefits paid to claimants under the subsidy program. In contrast, a control group of UI recipients who were not offered wage subsidy vouchers were paid an average of 2278 dollars in their initial spell of unemployment.

3. STANDARD DEVIATION, VARIATION AND PRECISION

The conventional parameterization for a normal random variable y is in terms of its mean μ and variance σ^2 . In this case, the pdf of y is given by

$$p(y|\mu, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2}(y - \mu)^2 \right\}.$$

We write $y \sim \text{Normal}(\mu, \sigma^2)$ to indicate this. The distribution of y is symmetric around μ with inflection points at $\mu \pm \sigma$.

The units of the variance are the square of the units of y , which can be difficult to interpret.¹ The *standard deviation* σ is the square root of the variance and has the same units as y . In R and Stan (but not JAGS—see below), the normal distribution is parameterized by the mean = μ and standard deviation $\text{sd} = \sigma$, not the variance. Thus, to draw 1000 normal random variables with $\mu = 1$ and $\sigma^2 = 4$, we would use `rnorm(1000, mean=1, sd=2)`.

In Bayesian analysis, it is often convenient to work with the reciprocal of the variance, called the *precision*. A precision of zero corresponds to infinite variance, while a variance of zero corresponds to infinite precision. High precision means low variance and vice versa. Let $\tau = 1/\sigma^2$ denote the precision of a $\text{Normal}(\mu, \sigma^2)$ random variable. In this parameterization, the pdf of y is given by

$$p(y|\mu, \tau) = (2\pi)^{-1/2} \tau^{1/2} \exp \left\{ -\frac{\tau}{2}(y - \mu)^2 \right\}.$$

The algebra is a bit easier because we don't have to deal with reciprocals of variances, but otherwise nothing has changed. This parameterization is used in JAGS: `y ~ dnorm(mu, tau)` represents a $\text{Normal}(\mu, 1/\tau)$ prior for y .

4. PRIOR AND POSTERIOR DISTRIBUTIONS FOR THE MEAN OF A NORMAL DISTRIBUTION

In this section, we derive the posterior distribution for the mean of a normal distribution with a known variance. We represent our prior distribution also using a normal distribution. We start with the simplest case: a single observation on a normal random variable. Using the proportionality trick (from the previous lecture), the proof requires only a little algebra.

¹For example, the mean of household income in the U.S. is currently approximately \$50,000 with a variance of roughly one billion. While not ideal (because of skewness in the income distribution), the standard deviation $\sqrt{1,000,000} \approx \$32,000$ is easier to interpret.

Proposition 1. If $y \sim \text{Normal}(\mu, \sigma^2)$ with prior distribution $\mu \sim \text{Normal}(\mu_0, \tau_0^2)$ and σ^2 is known, then the posterior distribution is $\mu|y \sim \text{Normal}(\mu_1, \tau_1^2)$ where

$$\mu_1 = \frac{\sigma^2 \mu_0 + \tau_0^2 y}{\sigma^2 + \tau_0^2}$$

$$\tau_1^2 = \frac{\sigma^2 \tau_0^2}{\sigma^2 + \tau_0^2}$$

Comment: This result is easier to remember and understand if stated in terms of precisions, rather than variances. Let $h = 1/\sigma^2$ denote the data precision, $k_0 = 1/\tau_0^2$ the prior precision, and $k_1 = 1/\tau_1^2$ the posterior precision. Then,

$$\mu_1 = \left(\frac{k_0}{k_1} \right) \mu_0 + \left(\frac{h}{k_1} \right) y$$

$$k_1 = k_0 + h$$

Note that the weights on the prior and the data value are non-negative ($k_0/k_1, h/k_1 \geq 0$) and sum to one ($k_0/k_1 + h/k_1 = 1$). In words, the posterior mean is a weighted average of the prior mean (with weight proportional to the prior precision) and the data value (with weight proportional to the data precision). The posterior precision is the sum of the prior precision and the data precision.

Proof. The model and prior are (omitting multiplicative terms that don't involve μ)

$$p(y|\mu) \propto \exp \left\{ -\frac{1}{2} h (y - \mu)^2 \right\}$$

$$p(\mu) \propto \exp \left\{ -\frac{1}{2} k_0 (\mu - \mu_0)^2 \right\},$$

so the posterior for μ is proportional to

$$\begin{aligned} p(\mu|y) &\propto p(\mu)p(y|\mu) \\ &\propto \exp \left\{ -\frac{1}{2} [k_0(\mu - \mu_0)^2 + h(y - \mu)^2] \right\} \\ &= \exp \left\{ -\frac{1}{2} [(k_0 + h)\mu^2 - 2(k_0\mu_0 + hy)\mu] \right\} \exp \left\{ -\frac{1}{2} (k_0\mu_0^2 + hy^2) \right\} \\ &\propto \exp \left\{ -\frac{1}{2} (k_0 + h) \left[\mu^2 - 2 \left(\frac{k_0\mu_0 + hy}{k_0 + h} \right) \mu + \left(\frac{k_0\mu_0 + hy}{k_0 + h} \right)^2 \right] \right\} \\ &= \exp \left\{ -\frac{1}{2} k_1 (\mu - \mu_1)^2 \right\}, \end{aligned}$$

where $k_1 = k_0 + h$ and $\mu_1 = (k_0\mu_0 + y)/k_1$. In the last line, we used the proportionality trick to drop one term that didn't depend on μ and add another. (This process is known as “completing the square.”) The last line is the kernel of a Normal distribution for μ with mean μ_1 and precision k_1 ,

$$p(\mu|y) = (2\pi)^{-1/2} k_1^{-1/2} \exp \left\{ -\frac{1}{2} k_1 (\mu - \mu_1)^2 \right\}.$$

□

Next, we consider a sample of size n . The previous result carries through except we now have n times as much data. The posterior mean is a weighted average of the prior mean (with weight equal to the prior precision) and the sample mean (with weight proportional to n times the data precision). The posterior precision is equal to the prior precision plus n times the data precision.

The easiest way to obtain this result is to use the fact that $\bar{y} \sim \text{Normal}(\mu, \sigma^2/n)$ and to treat $\bar{y}$ as the data value. This works because $\bar{y}$ is a sufficient statistic for μ when σ^2 is known. However, we will derive this result directly, which will be useful when we consider the case of unknown σ^2 later.

Proposition 2. *If $y_1, \dots, y_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$ random variables with prior distribution $\mu \sim \text{Normal}(\mu_0, \tau_0^2)$ and σ^2 is known, then the posterior distribution is $\mu|y_1, \dots, y_n \sim \text{Normal}(\mu_n, \tau_n^2)$ where*

$$\mu_n = \frac{\sigma^2 \mu_0 + n \tau_0^2 \bar{y}}{\sigma^2 + n \tau_0^2}$$

$$\tau_n^2 = \frac{\sigma^2 \tau_0^2}{\sigma^2 + n \tau_0^2}$$

Comment: Proposition 1 is a special case of Proposition 2 with $n = 1$. As before, it's easier to derive the result with precisions $h = 1/\sigma^2$ and $k_0 = 1/\tau_0^2$, in which case

$$\mu_n = \left(\frac{k_0}{k_n} \right) \mu_0 + \left(\frac{nh}{k_n} \right) \bar{y}$$

$$k_n = k_0 + nh$$

We see that information is additive, except we are now adding n data points, each with precision h . Alternatively, we may think of μ_n as a weighted average of $\mu_0, y_1, \dots, y_n$ with μ_0 having weight proportional to k_0 and each of the y_i 's having weight proportional to h .

Proof. If $y_1, \dots, y_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, then

$$\begin{aligned} p(y_1, \dots, y_n | \mu) &= \prod_{i=1}^n p(y_i | \mu) = \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} e^{-(y_i - \mu)^2 / 2\sigma^2} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right\} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{n(\bar{y} - \mu)^2 + (n-1)s^2}{2\sigma^2} \right\}, \end{aligned}$$

where we used

$$\begin{aligned} \sum_{i=1}^n (y_i - \mu)^2 &= \sum_{i=1}^n (y_i - \bar{y} + \bar{y} - \mu)^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 + 2(\bar{y} - \mu) \sum_{i=1}^n (y_i - \bar{y}) + n(\bar{y} - \mu)^2 \\ &= (n-1)s^2 + n(\bar{y} - \mu)^2. \end{aligned}$$

Here, $\bar{y}$ is the sample mean and s^2 the sample variance,

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

In terms of the precision $h = 1/\sigma^2$ and dropping multiplicative terms that don't involve μ ,

$$p(y_1, \dots, y_n | \mu) \propto \exp \left\{ -\frac{1}{2} n h (\bar{y} - \mu)^2 \right\}.$$

As before the prior is $p(\mu) \propto \exp\{-k_0(\mu - \mu_0)^2/2\}$, so the posterior is

$$\begin{aligned} p(\mu | y_1, \dots, y_n) &\propto p(\mu) p(y_1, \dots, y_n | \mu) \\ &\propto \exp \left\{ -\frac{1}{2} [k_0(\mu - \mu_0)^2 + n h (\bar{y} - \mu)^2] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} (k_0 + n h) \left[\mu^2 - 2 \left(\frac{k_0 \mu_0 + n h \bar{y}}{k_0 + n h} \right) \mu \right] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} k_n (\mu - \mu_n)^2 \right\} \end{aligned}$$

where $k_n = k_0 + n h$ and $\mu_n = (k_0 \mu_0 + n h \bar{y})/k_n$. This is the kernel of a Normal distribution for μ with mean μ_n and precision k_n ,

$$p(\mu | y_1, \dots, y_n) = (2\pi)^{-1/2} k_n^{-1/2} \exp \left\{ -\frac{1}{2} k_n (\mu - \mu_n)^2 \right\}.$$

Thus, $\mu | y_1, \dots, y_n \sim \text{Normal}(\mu_n, \tau_n^2)$ where $\tau_n^2 = 1/k_n^2$. □

5. INTERPRETATIONS OF THE POSTERIOR MEAN

The posterior mean is a *point estimate* of μ that combines the information from the prior with that from the data $y_1, \dots, y_n$. Here are three ways of understanding how the process of Bayesian updating works.

Compromise between μ_0 and $\bar{y}$. The posterior mean μ_n is a weighted average of the prior mean μ_0 and the data mean $\bar{y}$,

$$\mu_n = \lambda \mu_0 + (1 - \lambda) \bar{y},$$

where the weights are

$$\begin{aligned} \lambda &= \frac{k_0}{k_0 + n h} \propto k_0 \\ 1 - \lambda &= \frac{n h}{k_0 + n h} \propto n h \end{aligned}$$

The weights $(\lambda, 1 - \lambda)$ are non-negative, sum to one, and are proportional to the precisions $(k_0, n h)$. The weight on the prior mean decreases as $n \rightarrow \infty$ with the prior information eventually being overwhelmed by the data.

Adjustment of the prior μ_0 toward the data $\bar{y}$. The preceding equation can be rewritten

$$\mu_n = \mu_0 + (1 - \lambda)(\bar{y} - \mu_0),$$

showing that Bayesian updating shift the prior μ_0 toward $\bar{y}$ to obtain the posterior. The adjustment factor $1 - \lambda$ is between zero and one, so it's only a partial adjustment. The larger the number of data points n , the larger the adjustment.

Shrink the data mean toward the prior μ_0 . Alternatively, we can think of the posterior as a way of shrinking noisy data toward the prior,

$$\mu_n = \bar{y} - \lambda(\bar{y} - \mu_0).$$

The shrinkage is always in the opposite direction of $\bar{y} - \mu_0$; if $\bar{y} > \mu_0$, then $\mu_n < \bar{y}$, while if $\bar{y} < \mu_0$, then $\mu_n > \bar{y}$. Only in the happy (and unlikely) event that $\bar{y} = \mu_0$ does $\mu_n = \bar{y}$.

6. PREDICTIVE DISTRIBUTIONS

In the Bayesian paradigm, the statement “ y has a normal distribution with mean μ ” is made *conditional* upon the unknown parameter μ (and, implicitly, conditional upon σ^2 as well). The marginal distribution of y is given by

$$p(y) = \int p(y|\mu) p(\mu) d\mu.$$

This integral can be difficult to solve, but when the form of the posterior is known, we can obtain an expression for $p(y)$ by rearranging Bayes' Theorem,

$$p(y) = \frac{p(\theta) p(y|\theta)}{p(\theta|y)}.$$

This works in both the Normal and Binomial models.

Example 1. If $\theta \sim \text{Beta}(a, b)$ and $y|\theta \sim \text{Binomial}(n, \theta)$, then

$$p(y) = \frac{\frac{1}{B(a, b)} \theta^{a-1} (1 - \theta)^{b-1} \binom{n}{y} \theta^y (1 - \theta)^{n-y}}{\frac{1}{B(a + y, b + n - y)} \theta^{a+y-1} (1 - \theta)^{b+n-y-1}} = \binom{n}{y} \frac{B(a + y, b + n - y)}{B(a, b)},$$

which is known as the Beta-Binomial distribution. It has support on the integers $0, 1, \dots, n$.

Example 2. The same method of proof can be used to show that the marginal distribution of y is Normal when $\mu \sim \text{Normal}(\mu_0, \tau_0^2)$ and $y|\mu \sim \text{Normal}(\mu, \sigma^2)$ (with σ^2 known). We will avoid the algebra, but it's easy to obtain the marginal mean of y using the tower property of conditional expectations,

$$\mathbb{E}(y) = \mathbb{E}[\mathbb{E}(y|\mu)] = \mathbb{E}(\mu) = \mu_0,$$

and the variance using Eve's Law,

$$\mathbb{V}(y) = \mathbb{E}[\mathbb{V}(y|\mu)] + \mathbb{V}[\mathbb{E}(y|\mu)] = \mathbb{E}(\sigma^2) + \mathbb{V}(\mu) = \sigma^2 + \tau_0^2.$$

Thus, the marginal distribution of y (not conditional upon μ) can be shown² to be $\text{Normal}(\mu_0, \sigma^2 + \tau_0^2)$. This is called the prior predictive distribution of y . If you were asked to predict the value of

²A heuristic argument that the distribution is normal is that $y = \mu + \epsilon$ where $\epsilon \sim \text{Normal}(0, \sigma^2)$ independent of μ , so y is the sum of independent normal random variables.

y before seeing any data, you would need to take into account uncertainty about μ as well as the variability of y around μ .

Suppose now you've seen some data $y_1, \dots, y_n$ and are asked to predict the value of, say, another draw y_{n+1} from the same process. Conditional upon μ , the next draw y_{n+1} is independent of the previous draws. However, having observed the data, we update our beliefs to $\mu|y_1, \dots, y_n \sim \text{Normal}(\mu_n, \tau_n^2)$. Applying the same arguments used to obtain the prior predictive distribution, we conclude that $y_{n+1}|y_1, \dots, y_n \sim \text{Normal}(\mu_n, \sigma^2 + \tau_n^2)$. This is called the *posterior predictive distribution* of y .

7. DIGRESSION: INFORMATION AND LIKELIHOOD

In the theory of maximum likelihood estimation, the concept of information plays a central role. Suppose the data y depend on a one-dimensional parameter θ through the likelihood function $p(y|\theta)$. We define the *information* in y as

$$\mathcal{I}(\theta) = -\mathbb{E}\left(\frac{\partial^2 \log p(y|\theta)}{\partial \theta^2}\right) = -\int \frac{\partial^2 \log p(y|\theta)}{\partial \theta^2} p(y|\theta) dy$$

The expectation is taken using $p(y|\theta)$ (i.e., the same value of θ at which the partial derivatives are evaluated). The information measures the curvature of the likelihood function at θ .

Example 3. Suppose $y \sim \text{Binomial}(n, \theta)$ where n is known. Then

$$\log p(y|\theta) = \log \binom{n}{y} + y \log(\theta) + (n - y) \log(1 - \theta).$$

The first two derivatives of the log likelihood function are

$$\begin{aligned} \frac{\partial \log p(y|\theta)}{\partial \theta} &= \frac{y - n\theta}{\theta(1 - \theta)} \\ \frac{\partial^2 \log p(y|\theta)}{\partial \theta^2} &= -\frac{n}{\theta(1 - \theta)} - \frac{(1 - 2\theta)(y - n\theta)}{\theta^2(1 - \theta)^2} \end{aligned}$$

Taking expectations gives

$$\mathcal{I}(\theta) = -\mathbb{E}\left(\frac{\partial^2 \log p(y|\theta)}{\partial \theta^2}\right) = \frac{n}{\theta(1 - \theta)}$$

since $\mathbb{E}(y|\theta) = n\theta$.

Example 4. Suppose $y \sim \text{Normal}(\mu, \sigma^2)$ where the parameter is $\theta = \mu$ (the mean of the distribution) and the variance σ^2 is treated as known. Then

$$\log p(y|\theta) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y - \theta)^2}{2\sigma^2}$$

so the first two partial derivatives of the log likelihood function are

$$\begin{aligned} \frac{\partial \log p(y|\theta)}{\partial \theta} &= \frac{y - \mu}{\sigma^2} \\ \frac{\partial^2 \log p(y|\theta)}{\partial \theta^2} &= -\frac{1}{\sigma^2} \end{aligned}$$

Taking expectations, we obtain

$$\mathcal{I}(\theta) = -\mathbb{E}\left(\frac{\partial^2 \log p(y|\theta)}{\partial \theta^2}\right) = \frac{1}{\sigma^2}.$$

The information depends upon the parameterization used. For example, suppose instead of using θ we used some other parameterization $\phi = g(\theta)$ with differentiable inverse. Then the information matrix in terms of ϕ is

$$\begin{aligned}\mathcal{I}(\phi) &= -\mathbb{E}\left(\frac{\partial^2 \log p(y|\phi)}{\partial \phi^2}\right) \\ &= -\mathbb{E}\left[\frac{\partial^2 \log p(y|\theta)}{\partial \theta^2} \left(\frac{\partial \theta}{\partial \phi}\right)^2 + \frac{\partial \log p(y|\theta)}{\partial \theta} \frac{\partial^2 \theta}{\partial \phi^2}\right] \\ &= \mathcal{I}(\theta) \left(\frac{\partial \theta}{\partial \phi}\right)^2\end{aligned}$$

The *invariance principle* for maximum likelihood states that if $\hat{\theta}$ is the MLE using the first parameterization, then $\hat{\phi} = g(\hat{\theta})$ is the MLE using the second parametrization. We get equivalent estimates regardless of which parameterization we use. We also get the same inferences if we use likelihood ratio statistics. Wald and Score tests can give different inferences, since they are based on linear approximations at different points along the likelihood function.

8. THE JEFFREYS' PRIOR

Bayes' argued that if you know nothing about the success probability in a binomial model, you should use a uniform prior. Jeffreys' argued that this is incorrect since it depends upon the particular parameterization chosen. For example, suppose instead of parameterizing the binomial in terms of the success probability θ , we used instead the logit of the success probability,

$$\phi = \log \frac{\theta}{1 - \theta}.$$

Let $g(\theta) = \log[\theta/(1 - \theta)]$ so $g^{-1}(\phi) = (1 + e^{-\phi})^{-1}$. A uniform prior over θ implies the following prior over ϕ :

$$p_{\phi}(\phi) = p_{\theta}(g^{-1}(\phi)) \left| \frac{dg^{-1}(\phi)}{d\phi} \right| = \frac{e^{\phi}}{(1 + e^{\phi})^2},$$

which is definitely not uniform. (See Figure 1.)

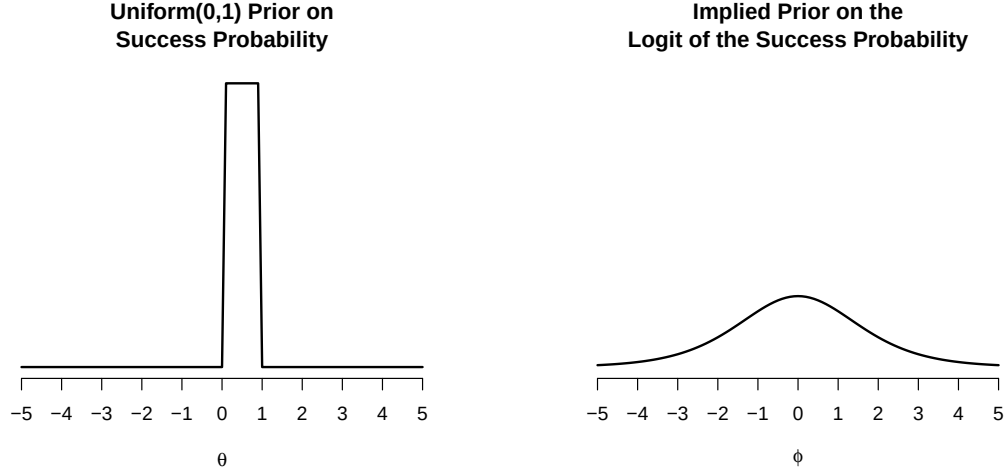


FIGURE 1. A Uniform(0, 1) prior on the success probability implies a symmetric, bell-shaped distribution on the logit of the success probability $\phi = \log[\theta/(1 - \theta)]$. It carries with it the information that the logit is between -5 and 5 with probability approximately 97.7%, corresponding to the success probability being between 0.007 and 0.993.

Jeffreys' argued that when we should prefer priors that are invariant to reparameterizations. This can be accomplished by using the prior

$$p(\theta) \propto \sqrt{\mathcal{I}(\theta)}.$$

It's easy to check that if $\phi = g(\theta)$, then we get $p(\phi) \propto \sqrt{\mathcal{I}(\phi)}$:

$$p_\phi(\phi) = p_\theta(g^{-1}(\phi)) \left| \frac{dg^{-1}(\phi)}{d\phi} \right| \propto \sqrt{\mathcal{I}(\theta)} \left| \frac{d\theta}{d\phi} \right| = \mathcal{I}(\phi).$$

Next, we derive the Jeffreys' prior for the Binomial and Normal models.

Example 5. If $y \sim \text{Binomial}(n, \theta)$, then the Jeffreys' prior is

$$p(\theta) \propto \sqrt{\frac{n}{\theta(1-\theta)}} \propto \theta^{-1/2}(1-\theta)^{-1/2}$$

which is a $\text{Beta}(1/2, 1/2)$ distribution.

Example 6. If $y \sim \text{Normal}(\mu, \sigma^2)$ with σ^2 known, then the Jeffreys' prior is

$$p(\mu) \propto \sqrt{\frac{1}{\sigma^2}} = \frac{1}{\sigma}.$$

This is a Uniform distribution on the real line, which is improper. We discuss it further in the next section.

9. IMPROPER PRIORS

In the binomial model, a Uniform(0, 1) distribution was used as a default, non-informative prior for the success probability θ . This worked because the support for θ was a finite interval. In the normal model, the support for μ could be the entire real line and, unfortunately, there is no uniform distribution on an infinite interval. However, we may use an *improper* prior

$$p(\mu) \propto 1$$

which does not have a finite integral.

Nonetheless, if we crank through the usual Bayesian calculations with this prior and data $y = (y_1, \dots, y_n)$, the posterior distribution is

$$\begin{aligned} p(\mu|y) &\propto p(\mu|\sigma^2) p(y|\mu, \sigma^2) \\ &\propto 1 \times \exp \left\{ -\frac{(n-1)s^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\} \\ &\propto \exp \left\{ -\frac{(\mu - \bar{y})^2}{2(\sigma^2/n)} \right\} \end{aligned}$$

which is the kernel of a Normal($\bar{y}$, σ^2/n) distribution. Thus, the posterior *does* have a finite integral and is a proper probability distribution.

Furthermore, the posterior can be thought of as the limit of the posterior arising from a sequence of priors with increasing variance. In particular, consider the sequence of priors $\mu \sim \text{Normal}(\mu_0, \tau_{0,k}^2)$ where $\tau_{0,k}^2 \rightarrow \infty$. (It's irrelevant what the prior mean μ_0 is, so long as it doesn't change.) Then, the sequence of posterior distributions are Normal(μ_k, τ_k^2) where, as $k \rightarrow \infty$ for fixed n ,

$$\begin{aligned} \frac{1}{\tau_k^2} &= \frac{1}{\tau_{0,k}^2} + \frac{n}{\sigma^2} \rightarrow \frac{n}{\sigma^2} \\ \mu_k &= \tau_k^2 \left(\frac{\mu_0}{\tau_{0,k}^2} + \frac{n\bar{y}}{\sigma^2} \right) \rightarrow \bar{y} \end{aligned}$$

We will see cases later where the limit of the posterior differs from the posterior obtained using an improper prior, but in this case everything works out.